HW0526

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16.1

16.1 In Examples 16.2 and 16.4, we presented the linear probability and probit model estimates using an example of transportation choice. The logit model for the same example is $P(AUTO = 1) = \Lambda(\gamma_1 + \gamma_2 DTIME)$, where $\Lambda(\bullet)$ is the logistic *cdf* in equation (16.7). The logit model parameter estimates and their standard errors are

$$\underset{(se)}{\tilde{\gamma}_1 + \tilde{\gamma}_2 DTIME} = \underset{(0.7505)}{-0.2376} + \underset{(0.2064)}{0.5311} DTIME$$

a. Calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$.

b. Using the probit model results in Example 16.4, calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$. How does this result compare to the logit estimate? [*Hint*: Recall that Statistical Table 1 gives cumulative probabilities for the standard normal distribution.]

c. Using the logit model results, compute the estimated marginal effect of an increase in travel time of 10 minutes for an individual whose travel time is currently 30 minutes longer by bus (public transportation). Using the linear probability model results, compute the same marginal effect estimate. How do they compare?

d. Using the logit model results, compute the estimated marginal effect of a decrease in travel time of 10 minutes for an individual whose travel time is currently 50 minutes longer by driving. Using the probit results, compute the same marginal effect estimate. How do they compare?

(a) Using the logit estimates, the probability of a person choosing automobile transportation given that $DTIME = 1$ is $\Lambda(\tilde{\gamma}_1 + \tilde{\gamma}_2 DTIME) = 0.5729$

(b) Using the probit estimates, the probability of a person choosing automobile transportation given that $DTIME = 1$ is $\Phi(\tilde{\beta}_1 + \tilde{\beta}_2 DTIME) = 0.5931$. There is very little difference between the logit and probit predicted probabilities.

(c) Given that $DTIME = 3$, the marginal effect of an increase in $DTIME$ using the logit estimates is

$$\widehat{\frac{dp}{dx}} = \lambda(\tilde{\gamma}_1 + \tilde{\gamma}_2 30)\tilde{\gamma}_2 = 0.086537$$

The estimated linear probability model is in Example 16.2. The marginal effect is the estimated slope of 0.0703, which is a bit smaller than the logit estimate. But the important point is that for the linear probability model the marginal effect is constant and doesn't depend on the existing commute time.

(d) The logit estimate is $\widehat{\frac{dp}{dx}} = \lambda(\tilde{\gamma}_1 + \tilde{\gamma}_2 \times -5)\tilde{\gamma}_2 = 0.02642$.

The probit estimate is $\widehat{\frac{dp}{dx}} = \phi(\tilde{\beta}_1 + \tilde{\beta}_2 \times -5)\tilde{\beta}_2 = 0.035204$

16.4

16.4 In Example 16.3, we illustrate the calculation of the likelihood function for the probit model in a small example. In this exercise, we will repeat that example using logit instead of probit. The logit model for the same example is $P(y=1) = \Lambda(\gamma_1 + \gamma_2 x)$, where $\Lambda(\bullet)$ is the logistic *cdf* in equation (16.7). The maximum likelihood estimates of the parameters are $\tilde{\gamma}_1 + \tilde{\gamma}_2 x = -1.836 + 3.021x$. The maximized value of the log-likelihood function is -1.612.

a. Calculate the probability that $y = 1$ if $x = 1.5$, given the values of the maximum likelihood estimates.

b. Using the threshold 0.5 and the result in part (a), predict the value of y if $x = 1.5$, the first observation, given the values of the maximum likelihood estimates. Compare your prediction to the actual outcome $y = 1$ in the first observation.

c. Calculate the value of the likelihood function, illustrated in equation (16.14) but substituting equation (16.17) in place of $\Phi(\bullet)$ and using the given $N = 3$ data pairs, if the parameter values are $\gamma_1 = -1$ and $\gamma_2 = 2$. Compare this value to the value of the likelihood function evaluated at the maximum likelihood estimates. Which is larger?

d. For the logit model, the value of the likelihood function (16.14), with $\Lambda(\bullet)$ in place of $\Phi(\bullet)$, will always be between zero and one. True or false? Explain.

e. For the logit model, the value of the log-likelihood function (16.15), with $\Lambda(\bullet)$ in place of $\Phi(\bullet)$, will always be negative. True or false? Explain.

(a)

$$z = -1.836 + 3.021(1.5) = 2.6955$$

$$P(y=1|x=1.5) = \Lambda(2.6955) = \frac{1}{1+e^{-2.6955}} \approx 0.9368$$

(b)

因為估計機率 $0.9368 \geq 0.5$，我們預測 $\hat{y} = 1$。這與第一筆觀察值的實際結果（$y_1 = 1$）**完全吻合**。

(c)

概似函數 $L = P(y_1 = 1) \times P(y_2 = 1) \times P(y_3 = 0)$：

- $P_1 = \Lambda(-1 + 2 \times 1.5) = \Lambda(2) = 0.8808$
- $P_2 = \Lambda(-1 + 2 \times 0.6) = \Lambda(0.2) = 0.5498$
- $P(y_3 = 0) = 1 - \Lambda(-1 + 2 \times 0.7) = 1 - \Lambda(0.4) = 1 - 0.5987 = 0.4013$

$$L = 0.8808 \times 0.5498 \times 0.4013 \approx 0.1943$$

比較： 題目給定在 MLE 下的對數概似值為 -1.612，對應的極大化概似函數值為 $e^{-1.612} \approx 0.1995$。顯然 **MLE 的概似值 (0.1995) 更大**，這證明了最大概似估計法（MLE）確實能找到使樣本發生機率（概似函數）最大化的參數。

(d)

True。 概似函數是所有獨立樣本事件發生機率的連乘積。既然每一個單獨的機率都必然介於 0 到 1 之間，它們的乘積自然也嚴格介於 0 與 1 之間。

(e)

True。 對數概似函數是對上述介於 0 與 1 之間的概似函數取自然對數（$\ln(L)$）。數學上，任何介於 0 到 1 之間的正數，其自然對數必定為負數。

16.18

16.18 Mortgage lenders are interested in determining borrower and loan characteristics that may lead to delinquency or foreclosure. In the data file *lasvegas* are 1000 observations on mortgages for single family homes in Las Vegas, Nevada during 2008. The variable of interest is *DELINQUENT*, an indicator variable = 1 if the borrower missed at least three payments (90 + days late), but 0 otherwise. Explanatory variables are *LVR* = the ratio of the loan amount to the value of the property; *REF* = 1 if purpose of the loan was a "refinance" and = 0 if loan was for a purchase; *INSUR* = 1 if mortgage carries mortgage insurance, 0 otherwise; *RATE* = initial interest rate of the mortgage; *AMOUNT* = dollar value of mortgage (in \$100,000); *CREDIT* = credit score, *TERM* = number of years between disbursement of the loan and the date it is expected to be fully repaid, *ARM* = 1 if mortgage has an adjustable rate, and = 0 if mortgage has a fixed rate.

a. Estimate the linear probability (regression) model explaining *DELINQUENT* as a function of the remaining variables. Use White heteroskedasticity robust standard errors. Are the signs of the estimated coefficients reasonable?

b. Use logit to estimate the model in (a). Are the signs and significance of the estimated coefficients the same as for the linear probability model?

c. Compute the predicted value of *DELINQENT* for the 500th and 1000th observations using both the linear probability model and the logit model. Interpret the values.

d. Construct a histogram of *CREDIT*. Using both linear probability and logit models, calculate the probability of delinquency for *CREDIT* = 500, 600, and 700 for a loan of \$250,000 (*AMOUNT* = 2.5). For the other variables, let the loan to value ratio (*LVR*) be 80%, the initial interest rate is 8%, all indicator variables take the value 0, and *TERM* = 30. Discuss similarities and differences among the predicted probabilities from the two models.

e. Using both linear probability and logit models, compute the marginal effect of *CREDIT* on the probability of delinquency for *CREDIT* = 500, 600, and 700, given that the other explanatory variables take the values in (d). Discuss the interpretation of the marginal effect.

f. Construct a histogram of *LVR*. Using both linear probability and logit models, calculate the probability of delinquency for *LVR* = 20 and *LVR* = 80, with *CREDIT* = 600 and other variables set as they are in (d). Compare and contrast the results.

g. Compare the percentage of correct predictions from the linear probability model and the logit model using a predicted probability of 0.5 as the threshold.

h. As a loan officer, you wish to provide loans to customers who repay on schedule and are not delinquent. Suppose you have available to you the first 500 observations in the data on which to base your loan decision on the second 500 applications (501–1,000). Is using the logit model with a threshold of 0.5 for the predicted probability the best decision rule for deciding on loan applications? If not, what is a better rule?

(a)

主要變數預期係數符號應符合經濟直覺。風險越高，違約機率越高：

- LVR(+)：貸款佔房產價值比例越高，借款人自有資金越少，違約風險較高。
- RATE(+)：利率越高，還款壓力大，正向影響。
- CREDIT(-)：信用分數越高代表過去信用越好，預期符號應為顯著的負號。

從下表可看出上述主要變數的係數皆具顯著性且方向符合預期。

```
t test of coefficients:

              Estimate  Std. Error  t value  Pr(>|t|)    
(Intercept)  0.68849128  0.22850643   3.0130 0.0026525 ** 
lvr          0.00162385  0.00067523   2.4049 0.0163601 *  
ref         -0.05932367  0.02402561  -2.4692 0.0137098 *  
insur       -0.48158489  0.03036944 -15.8575 < 2.2e-16 ***
rate         0.03437614  0.00981944   3.5008 0.0004845 ***
amount       0.02376801  0.01445089   1.6447 0.1003398    
credit      -0.00044190  0.00020728  -2.1319 0.0332620 *  
term        -0.01261951  0.00355596  -3.5488 0.0004051 ***
arm          0.12832392  0.02769325   4.6338  4.07e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

(b)

```
Call:
glm(formula = delinquent ~ lvr + ref + insur + rate + amount + 
    credit + term + arm, family = binomial(link = "logit"), data = lasvegas)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)    
(Intercept)  1.665303   1.992157   0.836 0.403195    
lvr          0.013519   0.008199   1.649 0.099164 .  
ref         -0.525064   0.230491  -2.278 0.022725 *  
insur       -3.122105   0.216906 -14.394  < 2e-16 ***
rate         0.308656   0.082771   3.729 0.000192 ***
amount       0.217606   0.113769   1.913 0.055786 .  
credit      -0.003564   0.001933  -1.844 0.065232 .  
term        -0.133037   0.034597  -3.845 0.000120 ***
arm          1.405649   0.371355   3.785 0.000154 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 998.03  on 999  degrees of freedom
Residual deviance: 666.29  on 991  degrees of freedom
AIC: 684.29

Number of Fisher Scoring iterations: 5
```

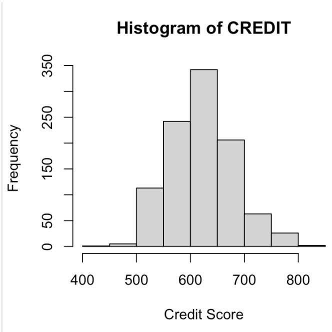
係數的正負號與顯著性與(a)大致相同。

(c)

```
> # LPM 預測
> predict(lpm_model, newdata = lasvegas[c(500, 1000), ])
      500      1000 
0.1827828 0.5785296 
> # Logit 預測 (務必加上 type="response" 才會輸出機率)
> predict(logit_model, newdata = lasvegas[c(500, 1000), ], type = "response")
      500      1000 
0.1335848 0.6265535 
```

LPM 的預測值在極端情況下可能會小於 0 或大於 1（這在機率上是不合理的）；而 Logit 模型透過轉換，確保預測值嚴格落在(0, 1)的區間內。分別的違約機率預測值如上表，可以看出第 500 筆相對第 1000 筆的違約機率預測都是比較低。

(d)



	credit	LPM_Prob	Logit_Prob
1	500	0.5532956	0.4967030
2	600	0.5091060	0.4086308
3	700	0.4649165	0.3260580

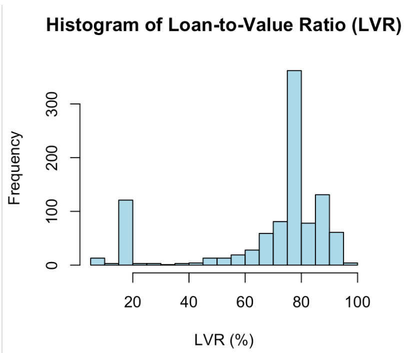
兩者在 credit 越高的情況下，違約機率預測值都會下降。但是在 Logit Model，由於非線性會下降更快。

(e)

	CREDIT	ME_LPM	ME_Logit
1	500	-0.0004418953	-0.0008910620
2	600	-0.0004418953	-0.0008613439
3	700	-0.0004418953	-0.0007832567

LPM 下，ME 不論在何等信用分數皆相同。而在 Logit Model 下，邊際效果會隨著信用分數的增加而遞減。

(f)



	LVR	Prob_LPM	Prob_Logit
1	20	0.4116749	0.2349087
2	80	0.5091060	0.4086308

LPM 的機率預估普遍較高，原因是 LPM 將變數對機率的影響假設為「直線等比例變化」，這使得 LPM 在處理極端情境（如非常低的 LVR 或非常高的 LVR）時，預測值可能會過度外推。相反地，Logit 模型採用了非線性的 Logistic S 型曲線。這種設計會將機率嚴格壓縮在 0 到 1 之間，並且在機率兩端的變化會趨於平緩。因此，Logit 模型在評估風險時，更能捕捉到金融實務中「極端值出現機率較難大幅變動」的特性，給出的預測通常比 LPM 更加穩健。

(g)

```
> cat("LPM 準確率 (Threshold=0.5):", round(accuracy_lpm * 100, 2), "%\n")
LPM 準確率 (Threshold=0.5): 85.8 %
> cat("Logit 準確率 (Threshold=0.5):", round(accuracy_logit * 100, 2), "%\n")
Logit 準確率 (Threshold=0.5): 85.4 %
```

兩者準確率並無出現太大差異，以 0.5 作為門檻的話。

(h)

「不，使用 0.5 作為預測機率的門檻並不是最好的決策法則。」在放貸實務中，我們必須考量「不對稱的錯誤成本（Asymmetric Cost of Errors）」。

第一型錯誤 (Type I Error)：預測不會違約而核貸，但客戶最終違約。這會造成極龐大的本金損失與法務催收成本。

第二型錯誤 (Type II Error)： 預測會違約而拒貸，但該客戶其實會按時還款。這造成的損失僅僅是「錯失的利息收入（機會成本）」。

由於「壞帳的資本損失」遠大於「錯失利息的機會成本」，金融機構不能容忍 50% 這麼高的違約風險。更好的決策法則是根據銀行的風險承受度，將門檻大幅下調。只要預測的違約機率超過這個嚴格的低門檻，就應當拒絕核貸，這樣才能有效攔截潛在的呆帳。